

```

// =====
// 1. KRIJIMI I DATABAZËS DHE KOLEKSIONEVE
// =====

// Kalon në databazën e projektit (nëse nuk ekziston, e krijon automatikisht)
use project_management_db;

// Krijon koleksionin e klientëve (pa validation)
db.createCollection("clients");

// Krijon koleksionin e përdoruesve me rregulla validimi (Schema Validation)
db.createCollection("users", {
  validator: {
    $jsonSchema: {
      bsonType: "object", // Dokumenti duhet të jetë objekt
      required: ["name", "email", "role"], // Fushat e detyrueshme
      properties: {
        name: { bsonType: "string", description: "Emri duhet të jetë tekst" },
        email: { bsonType: "string", pattern: "^.+@.+$", description: "Email duhet të jetë valid" },
        role: { enum: ["admin", "manager", "developer"], description: "Roli duhet të jetë një nga opsionet" }
      }
    }
  }
});

```

```
}  
});
```

```
// Krijon koleksionin e skuadrave (për lidhje Many-to-Many)
```

```
db.createCollection("teams");
```

```
// Krijon koleksionin e projekteve me Validation dhe Embedded Documents
```

```
db.createCollection("projects", {
```

```
  validator: {
```

```
    $jsonSchema: {
```

```
      bsonType: "object",
```

```
      required: ["title", "client_id", "status"],
```

```
      properties: {
```

```
        title: { bsonType: "string" },
```

```
        client_id: { bsonType: "objectId" }, // Referencë nga koleksioni clients
```

```
        status: { enum: ["active", "completed", "on_hold"] },
```

```
        start_date: { bsonType: "date" },
```

```
        milestones: { // Embedded Document (Array me objekte brenda projektit)
```

```
          bsonType: "array",
```

```
          items: {
```

```
            bsonType: "object",
```

```
            required: ["name", "deadline"],
```

```
            properties: {
```

```
              name: { bsonType: "string" },
```

```

        deadline: { bsonType: "date" },
        completed: { bsonType: "bool" }
    }
}
}
}
}
}
}
});

```

// Krijon koleksionin e detyrave (Tasks) me Validation

```

db.createCollection("tasks", {
    validator: {
        $jsonSchema: {
            bsonType: "object",
            required: ["title", "project_id", "assigned_to", "status"],
            properties: {
                title: { bsonType: "string" },
                project_id: { bsonType: "objectId" }, // One-to-Many: Një projekt ka
shumë task-e
                assigned_to: { bsonType: "objectId" }, // Referencë te përdoruesi
                status: { enum: ["todo", "in_progress", "done"] }
            }
        }
    }
});

```

```
}  
});  
  
// =====  
// 2. INSERTIMI I TË DHËNAVE (CREATE)  
// =====
```

```
// Shton klientët në sistem
```

```
db.clients.insertMany([  
  { _id: ObjectId("650c1f1a1c9d440000a1b1c1"), name: "Tech Corp", industry: "IT",  
    contact_email: "info@techcorp.com" },  
  { _id: ObjectId("650c1f1a1c9d440000a1b1c2"), name: "Alpha Finance", industry:  
    "Banking", contact_email: "contact@alpha.com" }  
]);
```

```
// Shton përdoruesit e sistemit
```

```
db.users.insertMany([  
  { _id: ObjectId("650c1f1a1c9d440000a1b2d1"), name: "Agim Hoxha", email:  
    "agim@test.com", role: "manager" },  
  { _id: ObjectId("650c1f1a1c9d440000a1b2d2"), name: "Bora Leka", email:  
    "bora@test.com", role: "developer" },  
  { _id: ObjectId("650c1f1a1c9d440000a1b2d3"), name: "Dritan Prifti", email:  
    "dritan@test.com", role: "developer" }  
]);
```

// Shton skuadrat (Lidhja Many-to-Many: Lista e members mban ID-të e përdoruesve)

```
db.teams.insertMany([
  { _id: ObjectId("650c1f1a1c9d440000a1b3e1"), name: "Backend Team", members:
[ObjectId("650c1f1a1c9d440000a1b2d2"),
ObjectId("650c1f1a1c9d440000a1b2d3")] },
  { _id: ObjectId("650c1f1a1c9d440000a1b3e2"), name: "Core Management",
members: [ObjectId("650c1f1a1c9d440000a1b2d1"),
ObjectId("650c1f1a1c9d440000a1b2d2")] }
]);
```

// Shton projektet me Embedded Documents (Milestones)

```
db.projects.insertMany([
  {
    _id: ObjectId("650c1f1a1c9d440000a1b4f1"),
    title: "E-Banking App",
    client_id: ObjectId("650c1f1a1c9d440000a1b1c2"),
    status: "active",
    start_date: new Date("2024-01-10"),
    milestones: [
      { name: "UI Design", deadline: new Date("2024-02-01"), completed: true },
      { name: "API Development", deadline: new Date("2024-04-01"), completed:
false }
    ]
  },
]);
```

```
{
  _id: ObjectId("650c1f1a1c9d440000a1b4f2"),
  title: "Server Migration",
  client_id: ObjectId("650c1f1a1c9d440000a1b1c1"),
  status: "on_hold",
  start_date: new Date("2024-03-15"),
  milestones: [ { name: "Backup Data", deadline: new Date("2024-03-20"),
completed: true } ]
}
]);
```

// Shton detyrat (Tasks) - Korrigjuar: pa shkronjën 'g' në ID

```
db.tasks.insertMany([
  { _id: ObjectId("650c1f1a1c9d440000a1b5a1"), title: "Setup Database",
project_id: ObjectId("650c1f1a1c9d440000a1b4f1"), assigned_to:
ObjectId("650c1f1a1c9d440000a1b2d2"), status: "done" },
  { _id: ObjectId("650c1f1a1c9d440000a1b5a2"), title: "Create Auth API",
project_id: ObjectId("650c1f1a1c9d440000a1b4f1"), assigned_to:
ObjectId("650c1f1a1c9d440000a1b2d3"), status: "in_progress" },
  { _id: ObjectId("650c1f1a1c9d440000a1b5a3"), title: "Configure AWS",
project_id: ObjectId("650c1f1a1c9d440000a1b4f2"), assigned_to:
ObjectId("650c1f1a1c9d440000a1b2d2"), status: "todo" }
]);
```

// =====

// 3. OPERACIONET CRUD (READ, UPDATE, DELETE)

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// =====
```

```
// READ: Gjen të gjitha task-et që janë në progres
```

```
db.tasks.find({ status: "in_progress" });
```

```
// READ: Gjen projektet e një klienti specifik
```

```
db.projects.find({ client_id: ObjectId("650c1f1a1c9d440000a1b1c2") });
```

```
// UPDATE: Ndryshon statusin e projektit nga on_hold në active
```

```
db.projects.updateOne(
```

```
  { _id: ObjectId("650c1f1a1c9d440000a1b4f2") },
```

```
  { $set: { status: "active" } }
```

```
);
```

```
// UPDATE (Complex): Shton një milestone të ri në fund të listës ekzistuese
```

```
db.projects.updateOne(
```

```
  { _id: ObjectId("650c1f1a1c9d440000a1b4f1") },
```

```
  { $push: { milestones: { name: "Frontend Integration", deadline: new Date("2024-05-15"), completed: false } } }
```

```
);
```

```
// DELETE: Fshin një task specifik nga sistemi
```

```
db.tasks.deleteOne({ _id: ObjectId("650c1f1a1c9d440000a1b5a3") });
```

```
// =====

// 4. AGGREGATION PIPELINES ($lookup)

// =====

// AGGREGATION 1: Bashkon projektet me klientët dhe llogarit numrin e
milestones

db.projects.aggregate([

  { $lookup: { from: "clients", localField: "client_id", foreignField: "_id", as:
"client_details" } }, // Join me clients

  { $unwind: "$client_details" }, // Kthen array-n në objekt

  { $project: { title: 1, status: 1, client_name: "$client_details.name",
total_milestones: { $size: "$milestones" } } } // Shfaq vetëm fushat që duam

]);

// AGGREGATION 2: Numëron sa task-e ka çdo përdorues dhe i rendit ata

db.users.aggregate([

  { $lookup: { from: "tasks", localField: "_id", foreignField: "assigned_to", as:
"user_tasks" } }, // Join me tasks

  { $project: { name: 1, role: 1, total_tasks_assigned: { $size: "$user_tasks" } } }, //
Kalkulon madhësinë e array-t

  { $sort: { total_tasks_assigned: -1 } } // Renditja zbritëse

]);

// =====

// 5. INDEXET (PËR PERFORMANCE)
```

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// =====
```

```
// Krijon index unik për email-in (Garanton që nuk ka dy përdorues me të njëjtin email)
```

```
db.users.createIndex({ email: 1 }, { unique: true });
```

```
// Krijon index për statusin e task-eve (Përshpejton kërkimet sipas statusit)
```

```
db.tasks.createIndex({ status: 1 });
```